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Unit – 5 : Session – 12 : SLO – 1 | ARM Addressing and Instruction Encoding

■ I. Multiple Choice

1. B) LDR R0, =0x40 — the literal-pool pseudo-instruction loads a constant.
2. B) Post-indexed — the base register is updated after the memory access.
3. C) Opcode + operands + condition — encoding covers all three fields.

■ II. Fill in the Blanks

1. before — pre-indexed addressing updates the base register before access.
2. LDR — literal constants are loaded with LDR Rd, =constant.
3. shift — scaled register offset applies a shift (e.g. LSL) to the index register.

■ III. True or False

1. TRUE.
2. FALSE — encoding identifies opcode, operands and condition fields, not just the opcode.
3. TRUE.

■ IV. Match the Following

- A) LDR — 3) Load data from memory.
B) STR — 1) Store register to memory.
C) Pre-indexed — 4) Base updated before access.
D) Literal LDR — 2) Load from constant pool.

■ V. Numerical Answers

1. 0x400C — $0x4000 + 12$.
2. ADD R4, R2, R1 → cond=1110 (AL), I=0, opcode=0100 (ADD), S=0, Rn=0010 (R2), Rd=0100 (R4), operand2=R1 → binary 1110 0000 1000 0010 0100 0000 0000 0001 = 0xE0824001.
3. 0xE2821005 → cond=1110, I=1 (immediate), opcode=0101 (ADC), S=0, Rn=R2, Rd=R1, operand2=#5 → ADC R1, R2, #5.